

Mark 3 ISO Installation, Operation & Maintenance Manual

Platform Alpha — document control cover (session insert)

Document number	FS-MARK3-IOM	Revision	C
Status	Current	Issue date	2025-11-10
Authority	Vendor (Flowserve)	Applies to	50-PA-101A; 50-PA-101B

Rev	Date	Description
B	2021-03-15	Initial issue
C	2025-11-10	Vibration limits split by bearing arrangement

This cover page is a session insert. Following pages are excerpted from the source publication listed in SOURCES.md.

FS-MARK3-IOM Rev C — Vibration limits by bearing arrangement

Revision C splits vibration acceptance by bearing arrangement. Arrangement 1 is stricter than Arrangement 2. This page does **not** state which installed pump has which arrangement — refer to SAP classification / MOC.

Configuration	Alarm (mm/s RMS)	Trip (mm/s RMS)	Bearing temp alarm/trip (°C)
Arrangement 1	4.5	7.1	85 / 95
Arrangement 2	5.6	9.0	85 / 95

Lubricant contamination is a leading cause of bearing distress. External lube oil supply from the shared LO unit must meet the LO unit manual targets.

Tags in this installation: 50-PA-101A, 50-PA-101B. Document: FS-MARK3-IOM Rev C (Current), issued 2025-11-10.



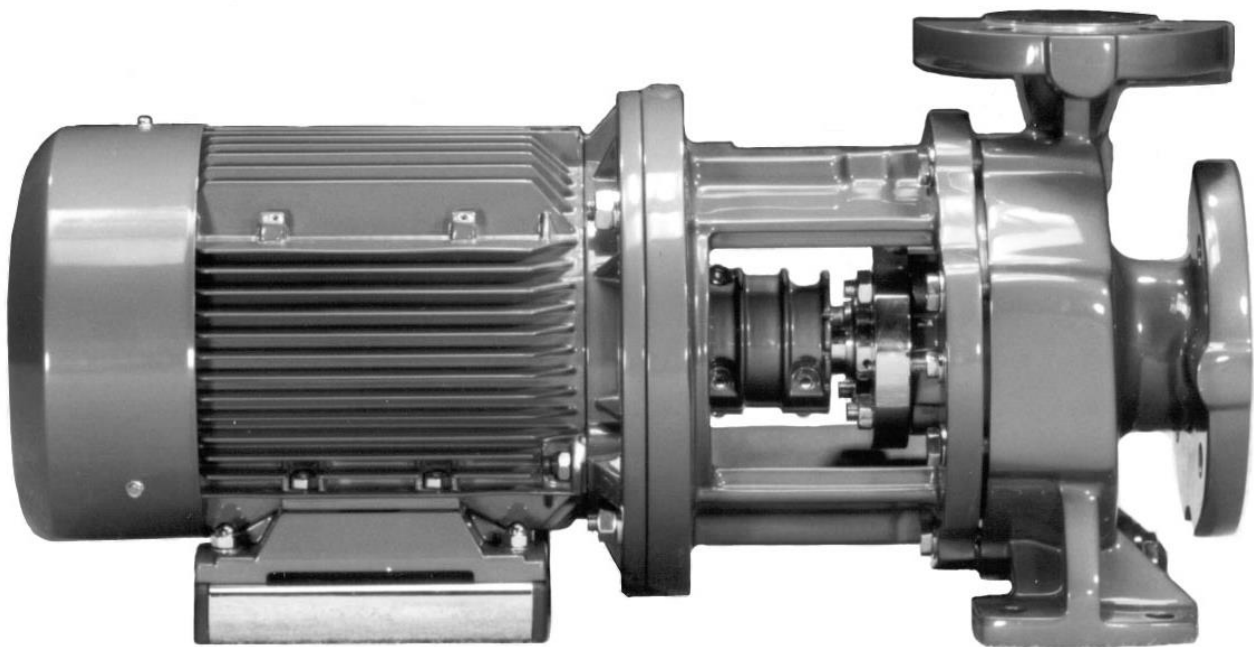
USER INSTRUCTIONS

Durco® Mark 3™ ISO close coupled

Close coupled, single stage, end suction, centrifugal,
chemical process pumps

PCN= 26999985 02-14 (E). Original instructions.

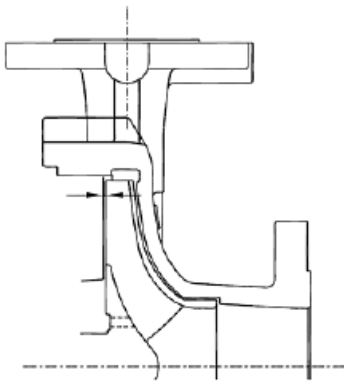
***Installation
Operation
Maintenance***



***These instructions must be read prior to installing,
operating, using and maintaining this equipment.***

Experience In Motion

5.2.2 Setting rear clearance on ISO-M reverse vane impeller (RV)



Clearance settings:

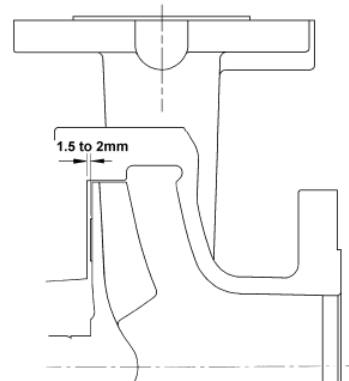
Impeller diameter		Motor frame size							
		80	90	100	112	132	160	180	200
Up to 210 mm	Clearance (mm)	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	Notches	7	7	7	7	7	8	8	8
211 to 315 mm	Clearance (mm)	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
	Notches	8	8	8	8	8	9	9	9

Reverse vane impellers are set off the cover [1220]. This allows the impeller [2200] to be set without the casing [1100].

- Disconnect the muff coupling [7120] and clean up the bores.
- Clean motor shaft and stub shaft and deburr where necessary.
- Replace muff coupling [7120], ensuring that the grub screw locates in the stub shaft.
- The motor end coupling bolts [6570.2] should be slacker than the pump end coupling bolts [6570.4] so that the coupling and stub shaft can be rotated relative to the motor shaft.
- The motor shaft should be prevented from rotating by using a C-spanner located in the keyway (where possible) or by locking the fan end of the motor [8100].
- With the casing [1100] removed, rotate the coupling [7120] until the impeller contacts the cover [1220]. This is the zero clearance position or datum for setting the rear clearance.
- Mark the bracket with a pen and, whilst preventing the motor shaft rotating, turn the coupling [7120] in the opposite direction by the recommended number of notches as indicated in the table. For the remaining steps, take care not to rotate the pump shaft relative to the motor shaft.
- Carefully loosen and back off the grub screw and tighten the coupling bolts [6570.4], ensuring that the gap is equal between the coupling halves [7120].
- Torque the screws to the specified values:
M8 – 30 Nm (22 lbf•ft)
M10 – 58 Nm (43 lbf•ft)

- Push the impeller [2200] axially towards the cover [1220] to close the motor bearing axial play.
- Ensure the impeller rear clearance is as indicated in the table using a feeler gauge.
- Check that the shaft can turn freely without binding.
- If a cartridge seal is fitted it should be reset at this point.

5.2.3 Setting rear clearance on ISO-RM recessed impeller



Recessed open impellers are set off the cover [1220]. This allows the impeller [2200] to be set without the casing [1100].

- Disconnect the muff coupling [7120] and clean up the bores. Ensure the casing [1100] is not fitted.
- Clean motor shaft and stub shaft and deburr where necessary.
- Replace muff coupling [7120], ensuring that the grub screw locates in the stub shaft.
- The motor end coupling bolts [6570.2] should be slacker than the pump end coupling bolts [6570.4] so that the coupling and stub shaft can be rotated relative to the motor shaft.
- The motor shaft should be prevented from rotating by using a C-spanner located in the keyway (where possible) or by locking the fan end of the motor [8100].
- With the casing [1100] removed, rotate the coupling [7120] until the rear clearance is 1.5 to 2 mm (0.06 to 0.08 in) as illustrated. This is the setting position. For the remaining steps, take care not to rotate the pump shaft relative to the motor shaft.
- Carefully loosen and back off the grub screw and tighten the coupling bolts [6570.4], ensuring that the gap is equal between the coupling halves [7120].
- Torque the screws to the specified values:
M8 – 30 Nm (22 lbf•ft)
M10 – 58 Nm (43 lbf•ft)
- Check that the shaft can turn freely without binding.
- If a cartridge seal is fitted it should be reset at this point.

5.3 Direction of rotation

 **CAUTION** Briefly run or jog the motor and observe the direction of rotation of the coupling or motor fan. Ensure the pump is given the same direction of rotation as the pump direction arrow on the bracket.

 **CAUTION** If maintenance work has been carried out to the site's electricity supply, the direction of rotation should be re-checked as above in case the supply phasing has been altered.

5.4 Guarding

 Guarding is supplied fitted to the pump set. If this has been removed or disturbed ensure that all the protective guards are securely refitted.

5.5 Priming and auxiliary supplies

5.5.1 Filling and priming

 **CAUTION** Ensure inlet pipe and pump casing is completely full of liquid before starting continuous duty operation.

Priming may be carried out with an ejector, vacuum pump interceptor or other equipment, or by flooding from the inlet source.

When in service, pumps using inlet pipes with foot valves may be primed by passing liquid back from the outlet pipe through the pump.

5.5.2 Auxiliary supplies

 **CAUTION** Ensure all electrical, hydraulic, pneumatic, sealant and lubrication systems (as applicable) are connected and operational.

5.6 Starting the pump

-  **CAUTION** Ensure flushing and or heating liquid supplies are turned ON, before starting pump.
- CLOSE the outlet valve.
- OPEN all inlet valves.
- Prime the pump.
- Start motor and check the outlet pressure.
- If the pressure is satisfactory, slowly OPEN the outlet valve.
-  **CAUTION** Do not run the pump with the outlet valve closed for a period longer than 30 seconds.
- If NO pressure, or LOW pressure, STOP the pump. Refer to section 7, *Faults; causes and remedies* for fault diagnosis.

5.7 Running the pump

5.7.1 Pumps fitted with mechanical seal

Mechanical seals require no adjustment. Any slight initial leakage will stop when the seal is run in.

Before pumping dirty liquids it is advisable, if possible, to run the pump in using clean liquid to safeguard the seal face.

 **CAUTION** External flush or quench should be started before the pump is run and allowed to flow for a period after the pump has stopped.

 **CAUTION** Never run a mechanical seal dry, even for a short time.

5.7.2 Normal vibration levels, alarm and trip

For guidance, pumps generally fall under a classification for rigid support machines within the International rotating machinery standards and the recommended maximum levels below are based on those standards.

 **CAUTION** Alarm and trip values for installed pumps should be based on the actual measurements (N) taken on the pump in the fully commissioned as new condition. Measuring vibration at regular intervals will then show any deterioration in pump or system operating conditions.

Vibration velocity – unfiltered	Horizontal pumps ≤ 15 kW mm/sec (in./sec) r.m.s.	> 15 kW mm/sec (in./sec) r.m.s.
Normal N	≤ 3.0 (0.12)	≤ 4.5 (0.18)
Alarm N x 1.25	≤ 3.8 (0.15)	≤ 5.6 (0.22)
Shutdown trip N x 2.0	≤ 6.0 (0.24)	≤ 9.0 (0.35)

5.7.3 Stop/start frequency

Pump sets are normally suitable for the number of equally spaced stop/starts per hour shown in the table below. Check capability of the driver and control/starting system before commissioning.

Motor rating kW (hp)	Maximum stop/starts per hour
Up to 15 (20)	15
Between 15 (20) and 90 (120)	10
Above 90 (120)	6

Where duty and standby pumps are installed it is recommended that they are run alternately every week.

5.8 Stopping and shutdown

-  **CAUTION** Close the outlet valve, but ensure that the pump runs in this condition for no more than a few seconds.
- Stop the pump.

- c) Switch off flushing and or heating liquid supplies at a time appropriate to the process.

- d)  For prolonged shut-downs and especially when ambient temperatures are likely to drop below freezing point, the pump and any cooling and flushing arrangements must be drained or otherwise protected.

5.9 Hydraulic, mechanical and electrical duty

This product has been supplied to meet the performance specifications of your purchase order, however it is understood that during the life of the product these may change. The following notes may help the user decide how to evaluate the implications of any change. If in doubt contact your nearest Flowserve office.

5.9.1 Specific gravity (SG)

Pump capacity and total head in metres (feet) do not change with SG, however pressure displayed on a pressure gauge is directly proportional to SG. Power absorbed is also directly proportional to SG. It is therefore important to check that any change in SG will not overload the pump driver or over-pressurize the pump.

5.9.2 Viscosity

For a given flow rate the total head reduces with increased viscosity and increases with reduced viscosity. Also for a given flow rate the power absorbed increases with increased viscosity, and reduces with reduced viscosity. It is important that checks are made with your nearest Flowserve office if changes in viscosity are planned.

5.9.3 Pump speed

Changing pump speed effects flow, total head, power absorbed, $NPSH_R$, noise and vibration. Flow varies in direct proportion to pump speed, head varies as speed ratio squared and power varies as speed ratio cubed. The new duty, however, will also be dependent on the system curve. If increasing the speed, it is important therefore to ensure the maximum pump working pressure is not exceeded, the driver is not overloaded, $NPSH_A > NPSH_R$, and that noise and vibration are within local requirements and regulations.

5.9.4 Net positive suction head ($NPSH_A$)

$NPSH$ available ($NPSH_A$) is a measure of the head available in the pumped liquid, above its vapour pressure, at the pump suction branch.

$NPSH$ required ($NPSH_R$) is a measure of the head required in the pumped liquid, above its vapour pressure, to prevent the pump from cavitating. It is

important that $NPSH_A > NPSH_R$. The margin between $NPSH_A > NPSH_R$ should be as large as possible.

If any change in $NPSH_A$ is proposed, ensure these margins are not significantly eroded. Refer to the pump performance curve to determine exact requirements particularly if flow has changed.

If in doubt please consult your nearest Flowserve office for advice and details of the minimum allowable margin for your application.

5.9.5 Pumped flow

Flow must not fall outside the minimum and maximum continuous safe flow shown on the pump performance curve and or data sheet.

6 MAINTENANCE

6.1 General

 It is the plant operator's responsibility to ensure that all maintenance, inspection and assembly work is carried out by authorized and qualified personnel who have adequately familiarized themselves with the subject matter by studying this manual in detail. (See also section 1.6.2.)

Any work on the machine must be performed when it is at a standstill. It is imperative that the procedure for shutting down the machine is followed, as described in section 5.8.

On completion of work all guards and safety devices must be re-installed and made operative again.

Before restarting the machine, the relevant instructions listed in section 5, *Commissioning, start-up, operation and shut down* must be observed.

Oil and grease leaks may make the ground slippery. Machine maintenance must always begin and finish by cleaning the ground and the exterior of the machine.

If platforms, stairs and guard rails are required for maintenance, they must be placed for easy access to areas where maintenance and inspection are to be carried out. The positioning of these accessories must not limit access or hinder the lifting of the part to be serviced.

When air or compressed inert gas is used in the maintenance process, the operator and anyone in the vicinity must be careful and have the appropriate protection.